AI Response Quality Evaluation System

Frontend Technical Architecture & Comprehensive System Documentation

Infosys Springboard Internship Project | Author: Athul-042 | October 2026

1. Executive Summary & Objective

The AI Response Quality Evaluation System is an enterprise-grade AI observability and validation platform designed to audit AI-generated or user-submitted responses against authoritative grounding evidence.

Unlike traditional conversational agents that blindly generate text, this system functions as an independent quality assurance judge. It decomposes responses into atomic factual assertions, executes semantic vector retrieval over knowledge documents using FAISS, evaluates claims using specialized LLM judges, and produces transparent audit verdicts with explicit scientific uncertainty quantification.

2. Core Problems Addressed

- Hallucination Risk: Detects fabricated dates, numbers, citations, or synthetic entities.
- Topical Drift: Identifies verbose responses that fail to address the core user prompt.
- Factual Contradiction: Flags assertions that directly conflict with verified ground truth.
- Critical Omissions: Analyzes missing prerequisites, caveats, or safety warnings.

3. Technology Stack

- Frontend Framework: React 19, TypeScript, and Vite 8.3 for fast build and strict type safety.
- Styling & Theme: Tailwind CSS v3.4 enterprise SaaS theme (Deep Navy #0F172A, Blue #2563EB).
- Data Visualizations: Recharts for interactive Area, Line, Bar, and Donut telemetry charts.
- Navigation & Icons: React Router v7 across 8 dedicated pages with Lucide React enterprise icons.
- Vector Engine & Storage: FAISS vector search simulation with all-MiniLM-L6-v2 embeddings and localStorage.

Multi-Agent Architecture & Mathematical Scoring

Section 2: The 5 Logical Agents, Scoring Formula, and Safety Veto

1. The Five Logical Evaluation Agents

- Agent 1: Supervisor / Orchestrator: Manages workflow queues, breaks text into atomic claims, coordinates FAISS evidence retrieval, and sc
- Agent 2: Relevance Judge (Weight: 20%): Checks prompt-response intent alignment without drift (Score: 0.0 to 5.0).
- Agent 3: Accuracy Judge (Weight: 30%): Cross-examines decomposed claims against retrieved evidence passages (Score: 0.0 to 5.0).
- Agent 4: Hallucination Sentinel (Weight: 25%): Identifies ungrounded or contradicted claims across 4 scientific categories (Score: 0.0 to 5.0)
- Agent 5: Completeness Judge (Weight: 25%): Detects omissions, missing dimensions, and unaddressed requirements (Score: 0.0 to 5.0).
- Verdict Aggregator Engine: Synthesizes all dimension scores into a 0-100 composite score and applies verdict gates.

2. The 4 Scientific Claim Categories

- Supported (Green): The claim is directly confirmed by retrieved reference passages.
- Unsupported (Amber): The available reference text does not mention the fact (missing evidence). Crucial rule: Missing evidence is not assur
- Contradicted (Red): Reliable reference documentation directly conflicts with or disproves the claim.
- Inconclusive (Slate): The statement is ambiguous, vague, or lacks context for high-confidence attribution.

3. Composite Formula & Non-Linear Safety Veto

The composite score is calculated using the weighted equation:

Score = (0.20 * Relevance + 0.30 * Accuracy + 0.25 * Hallucination + 0.25 * Completeness) * 20

Verdict Gates: PASS (Score >= 80%), NEEDS IMPROVEMENT (60% to 79%), FAIL (< 60%).

CRITICAL SAFETY VETO: If any factual claim is classified as Contradicted in high-risk domains (medical, financial, compliance), the system issues an automatic FAIL verdict regardless of the weighted average.

Detailed Page-by-Page Specifications

Section 3: Overview of All 8 Implemented Enterprise Frontend Pages

Overview of the 8 Dedicated Routes

- 1. Dashboard (/dashboard): Executive monitoring center with 4 KPI cards, evaluation volume area chart, verdict distribution donut, dimensio
- 2. Evaluate Response (/evaluate): Primary working form. Takes question, AI response, gold reference, and document upload. Includes 3 pre
- 3. Evaluation Results (/evaluations/:id): Deep-dive audit report. Displays large overall score (0-100), verdict badge, 4 dimension score cards,
- 4. Evaluation History (/history): Searchable audit archive with sorting, pagination, multi-filtering (verdict, score, hallucination status), side-by-s
- 5. Agent Architecture (/agents): Interactive topology map illustrating the end-to-end evaluation flow. Clicking any agent opens prompt templa
- 6. Knowledge Base (/knowledge-base): Manages trusted reference documents. Displays document chunks, vector previews, upload modal,
- 7. Analytics & Reports (/analytics): Analytics workspace with 6 Recharts graphs analyzing evaluation throughput, dimension trends, claim dis
- 8. Batch Evaluation (/batch-evaluation): Dataset auditing at scale. Supports CSV upload, downloadable sample CSV template, schema valid

Demonstration Guide & Mentor Presentation Script

Section 4: Step-by-Step Walkthrough for Infosys Springboard Evaluation

1. Recommended Presentation Flow for Evaluators

- Step 1: Dashboard Overview: Show the KPI cards (Total Audits, Average Score, Hallucination Rate) and explain that the system monitors qu
- Step 2: Start an Audit on Evaluate Page: Navigate to /evaluate. Click "Load Example Presets" and choose the "Hallucination Case (Gemini S
- Step 3: Watch the 9-Stage Progress Tracker: Click "Evaluate Response". Explain to the mentor how the system coordinates validation, claim
- Step 4: Analyze Findings on Results Page: Show the composite score (48/100, FAIL). Expand Claim #1 and Claim #3 to show the red CONT
- Step 5: Explain the Scientific Uncertainty Principle: Emphasize that missing evidence is labeled Unsupported (amber), not falsely called a lie
- Step 6: Showcase Knowledge Base & FAISS Playground: Navigate to /knowledge-base. Enter a query in the Semantic Retrieval box to show
- Step 7: Demonstrate Batch Dataset Evaluation: Navigate to /batch-evaluation. Click "Load Sample Dataset" and run the batch audit to prove

2. Summary Statement for Technical Mentors

"Our project provides an end-to-end, explainable, and factually grounded evaluation platform for AI applications. By leveraging a 5-agent LLM-as-a-Judge architecture combined with FAISS dense vector retrieval, we eliminate black-box scoring and provide transparent claim-by-claim verification suitable for enterprise deployment."